

Meeting Summarizer

Technical Report

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Project: Meeting Summarizer (Project 14)
Dataset: SAMSum
Platform: Apple M4 Pro · 24 GB UMA · MPS / BF16
Final Result: ROUGE-L **40.12** (target ≥ 40)

Project Outcome. The project solves abstractive dialogue summarization on the SAMSum benchmark using fine-tuned seq2seq transformers and an interactive Streamlit demo. The best system is `facebook/bart-base` trained on speaker-tagged dialogues and decoded with beam search (Config D27: `num_beams=5`, `length_penalty=1.33`), achieving **ROUGE-L 40.12** on the full 819-example test set.

1 Executive Summary and Success Criteria

This project builds an end-to-end **abstractive dialogue summarization system** for multi-speaker conversations. The system takes a chat-style conversation as input and returns a concise summary, along with optional structured outputs in the demo interface.

The evaluation target was explicit: **ROUGE-L ≥ 40.0** on the 819-example SAMSum test set. The final system achieved **ROUGE-L 40.12**, meeting the target using a mid-size model trained locally on Apple Silicon.

Three model families were evaluated: `facebook/bart-base`, `t5-small`, and `google/pegasus-cnn_dailymail`.

2 Problem Definition and Scope

2.1 Task

Given a multi-turn, multi-speaker dialogue, generate a concise abstractive summary that preserves the main facts, decisions, and speaker attribution.

Example input:

```
Amanda: I baked cookies. Do you want some?  
Jerry: Sure!  
Amanda: I'll bring you tomorrow :-)  
Jerry: Thanks! Do you know how to make the lemon ones?  
Amanda: The biscuits?  
Jerry: Yeah.  
Amanda: I'll send you the recipe. It's easy!
```

Example output:

Amanda will bring Jerry cookies tomorrow and send him the recipe for lemon biscuits.

2.2 Why This Task Is Difficult

- Dialogues are informal, compressed, and often omit context.
- Correct summarization requires **speaker attribution**, not just topic extraction.
- The average compression ratio is about **5.2×**, so omission and hallucination are realistic risks.
- Pre-trained summarization checkpoints are usually trained on news or web text rather than conversation.

2.3 Success Criterion

Primary target: ROUGE-L ≥ 40.0 on the full SAMSum test split.

Achieved: ROUGE-L 40.12 using BART-base + speaker tags + tuned decoding.

3 Dataset and Data Quality

3.1 Dataset Overview

Property	Value
Source	knkarthick/samsum on Hugging Face
Reference	Gliwa et al., 2019
Domain	English messenger-style dialogues
License	CC BY-NC-ND 4.0
Task type	Abstractive dialogue summarization

Split	Dialogues
Train	14,731
Validation	818
Test	819
Total	16,368

3.2 Data Quality Evidence

- **Leakage check passed:** scripts/data_audit.py found zero overlap across train, validation, and test IDs.
- **Dialogue length:** mean 148.9 tokens, p99 525 tokens.
- **Summary length:** mean 28.7 tokens, max 94 tokens.
- **Speaker structure:** 73% of dialogues have exactly two speakers.
- **Training implication:** max_source_length=512 covers nearly all dialogues, so truncation affects only an edge case.

SAMSum Dataset — Dialogue/Summary Example

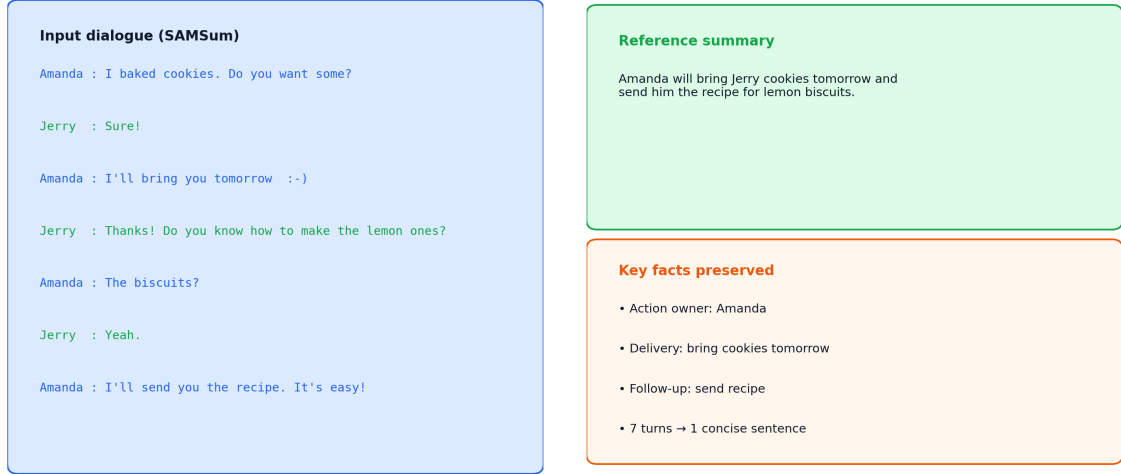


Figure 1: Representative SAMSum dialogue example used to illustrate the input structure and target summary style.

3.3 Dataset Risk

SAMSum is strong for controlled benchmarking, but it is **constructed rather than transcribed**. That means it is cleaner than real meeting transcripts and may overestimate performance on noisy production conversations.

4 Preprocessing and Baselines

4.1 Preprocessing Variants

Three preprocessing variants were evaluated because they directly affect the scoring categories for data handling and experiments:

- **with_speakers:** preserve speaker tags such as `Amanda:`.
- **no_speakers:** strip speaker tags to test the value of explicit attribution.
- **split_speakers:** use sliding-window segmentation for dialogues that exceed 512 tokens.

Data Preprocessing Pipeline — Three Tokenized Variants

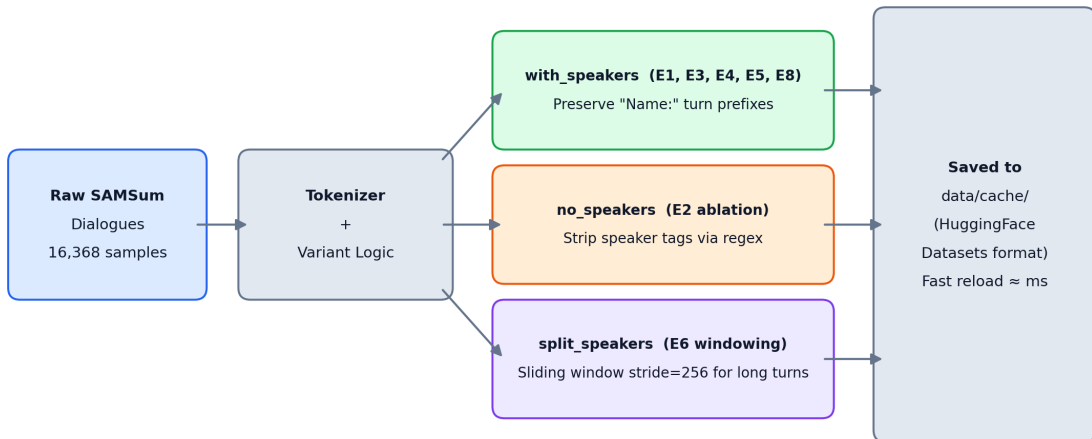


Figure 2: Preprocessing pipeline used to create the three training variants from raw SAMSum dialogues.

4.2 Zero-Shot Baselines

Model	ROUGE-1	ROUGE-2	ROUGE-L	Test Size
BART-base	27.34	8.87	19.89	100
T5-small	27.60	7.63	22.19	100
PEGASUS	1.85	0.00	1.60	100

These baselines establish two points: fine-tuning is necessary, and larger models do not automatically transfer well to dialogue summarization.

5 Models and Experiment Design

5.1 Model Choice

Model	Params	Role in project	Why included
facebook/bart-base	139M	Primary model	Strong encoder-decoder baseline for abstractive summarization
t5-small	60M	Compact comparison baseline	Tests a smaller text-to-text model under the same task
google/pegasus-cnn_dailymail	568M	Transfer stress test	Tests cross-domain transfer from news summarization

5.2 Experiment Matrix

ID	Change tested	ROUGE-L	Cost	Conclusion
E0	Zero-shot baselines	19.89–22.19	eval only	Fine-tuning is required
E1	BART vs T5 full fine-tuning	39.85 vs 31.95	72.4 vs 35.1 min	BART is clearly stronger
E2	Remove speaker tags	33.23	standard train	Speaker attribution is crucial
E3	Tune decoding hyperparameters	40.12	no retraining	Best final result
E4	Faithfulness evaluation	n/a	eval only	ROUGE alone is insufficient
E5	LoRA instead of full fine-tune	37.59	54.7 min	Efficient but lower quality
E6	Sliding-window handling	39.08	standard train	Extra complexity not justified
E7	PEGASUS transfer	1.15	1 epoch	Severe domain mismatch
E8	Longer training schedule	38.46	259.6 min	Slower and worse than default

6 Training Setup and Efficiency

6.1 Core Training Configuration

Setting	Value	Why it matters
Max source length	512	Covers nearly all dialogues
Max target length	128	Above the observed max summary length
Batch size	8	Safe on 24 GB UMA with BF16
Learning rate	5e-5	Standard fine-tuning configuration
Epochs	5	Best schedule among tested defaults
Early stopping	patience = 2	Reduces wasted computation
Default decoding	beam = 4, lp = 1.0	Training-time baseline
Best decoding	beam = 5, lp = 1.33	Final quality gain without re-training

6.2 Efficiency Summary

Configuration	Train time	Key metric	Efficiency takeaway
T5-small (E1)	35.1 min	ROUGE-L 31.95	Faster but well below target
BART-base (E1)	72.4 min	ROUGE-L 39.85	Best full fine-tune baseline
BART-base + D27 (E3)	retrain needed	not ROUGE-L 40.12	Best overall result
BART-base LoRA (E5)	54.7 min	ROUGE-L 37.59	Useful speed/quality compromise
BART extended (E8)	259.6 min	ROUGE-L 38.46	Much slower and worse

Training logs show post-training MPS memory near **16.96 GB**, which is consistent with the selected batch size and BF16 configuration on Apple Silicon.

7 Evaluation Methodology

7.1 Primary Metrics

- **ROUGE-1**: unigram overlap.
- **ROUGE-2**: bigram overlap.
- **ROUGE-L**: longest common subsequence; this is the **primary selection metric**.

All ROUGE values are reported as macro-averaged F-measures $\times 100$ over the full 819-example test set unless otherwise noted.

7.2 Supporting Evaluation

The report uses three supporting checks beyond ROUGE:

- **Bootstrap confidence intervals** for the main architecture comparison.
- **Faithfulness metrics** including entity hallucination, speaker preservation, and NLI entailment.

- **Manual error analysis** on 20 examples for qualitative failure inspection.

Metric	Value	Interpretation
Entity hallucination rate	10.1% (83/819)	Roughly 1 in 10 outputs invents an entity
Speaker preservation	75.5%	Most summaries preserve speaker identity
NLI faithfulness	0.308	Moderate entailment; useful but insufficient alone

8 Results and Best Model Selection

8.1 Main Results

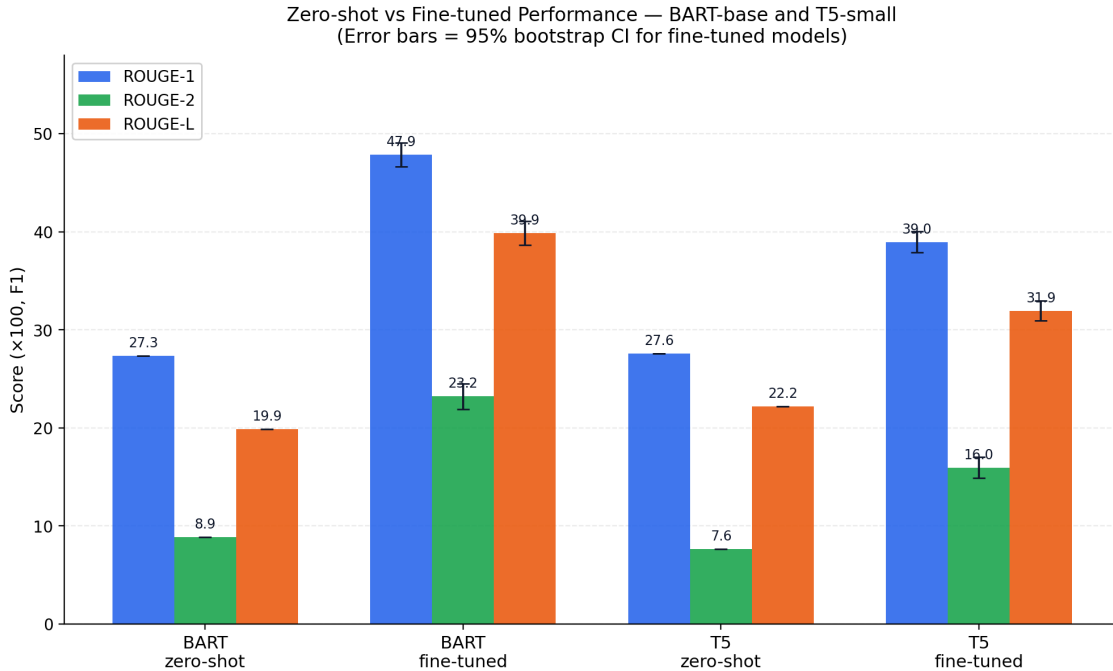


Figure 3: Zero-shot and fine-tuned model comparison with uncertainty on the fine-tuned systems.

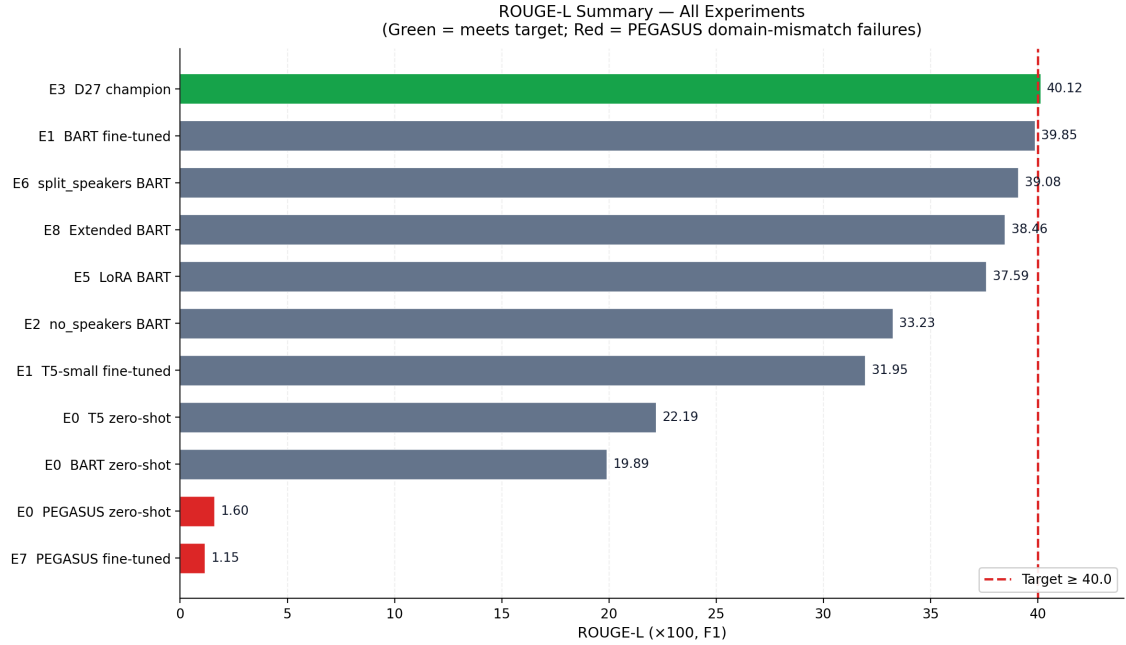


Figure 4: ROUGE-L across the reported experiments. The final system crosses the project target.

8.2 Key Findings

- **Architecture:** BART-base outperformed T5-small by **+7.90 ROUGE-L** after fine-tuning.
- **Speaker tags:** removing them reduced ROUGE-L from 39.85 to **33.23**, making this the most important preprocessing decision.
- **Decoding:** D27 improved ROUGE-L from 39.92 to **40.12** without retraining.
- **Efficiency:** LoRA retained useful quality (37.59) with less cost than full fine-tuning.
- **Longer training:** E8 was slower and worse, so extra epochs were not justified.

8.3 Best Model Selection

Property	Best configuration
Model	facebook/bart-base
Preprocessing	with_speakers
Decoding	num_beams=5, length_penalty=1.33 (D27)
ROUGE-1 / ROUGE-2 / ROUGE-L	48.48 / 23.55 / 40.12
Inference latency	200.6 ms/sample
Reason selected	Highest test ROUGE-L while remaining reproducible on local hardware

9 Error Analysis and Key Insights

9.1 Error Distribution

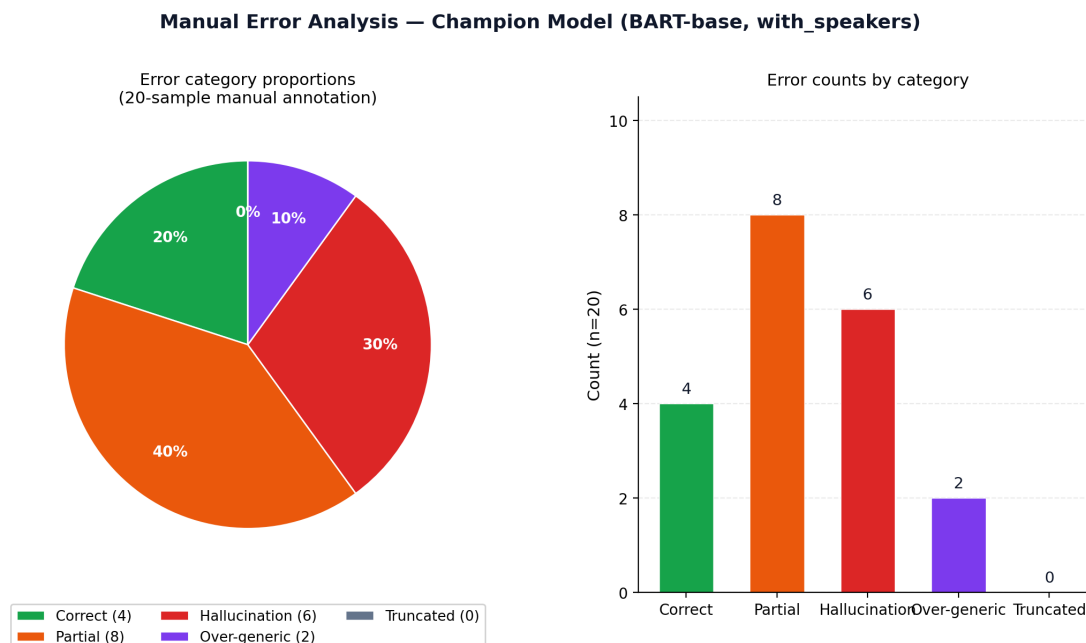


Figure 5: Manual error analysis for a 20-example sample from the champion model outputs.

Category	Count	Share
Correct	4	20%
Partial	8	40%
Hallucination	6	30%
Over-generic	2	10%
Truncated	0	0%

9.2 Representative Failure Cases

Failure type	Observed output	Why it matters
Hallucination	“Pogba scored the first goal of the season”	The dialogue discussed form, not a goal; this is fabricated content
Action inversion	“Randolph will buy earplugs for Maya”	The vocabulary is correct, but actor and recipient are reversed
Missing action detail	“Tom and Ben will meet at 2 pm”	Logistics are preserved, but the instruction to bring papers is lost

9.3 Three High-Value Insights

- **Speaker tags are the most impactful preprocessing choice.**
- **Decoding optimisation is free performance gain** once the model is trained.
- **Faithfulness remains the main weakness** even when ROUGE-L is strong.

10 Demo Application

10.1 What the Demo Shows

The Streamlit demo provides explicit evidence that the project is not only a training pipeline but also a usable application. The interface supports:

- dialogue input through a text area,
- model selection from available checkpoints,
- decoding controls such as beam width and length penalty,
- generated summary output,
- extracted action items,
- named entities,
- generation metadata including model name, device, and latency.

10.2 Screenshots

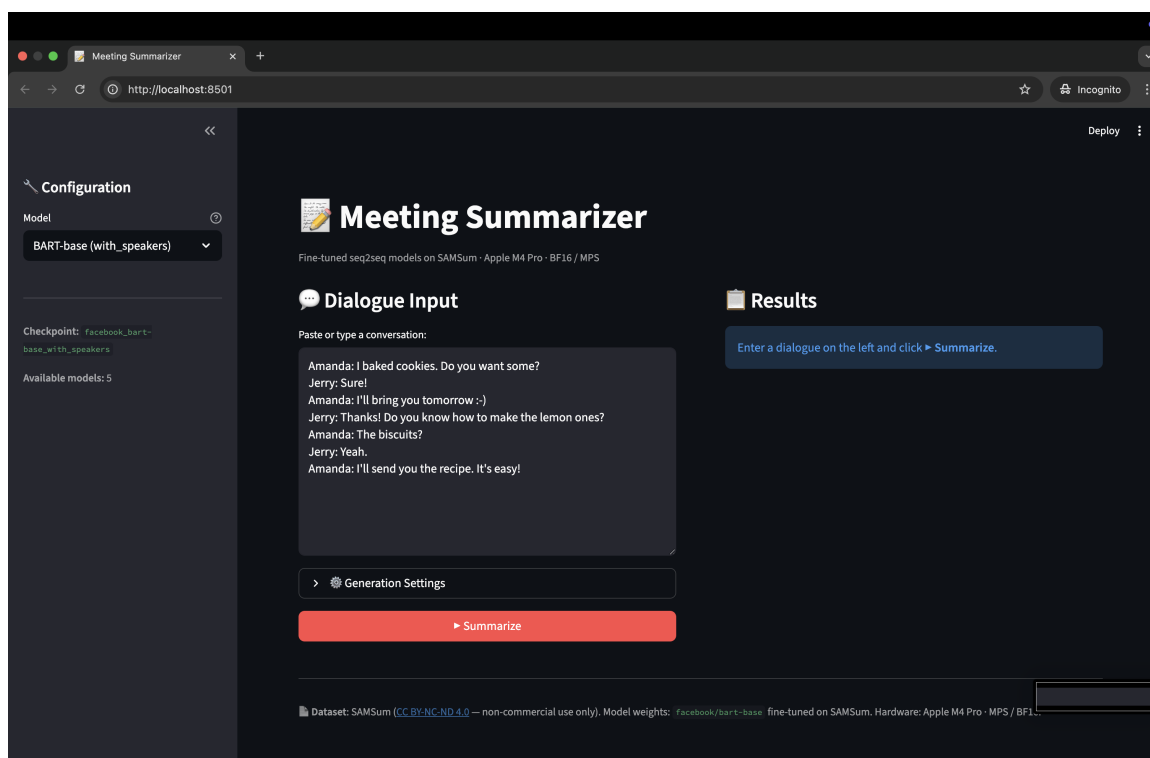


Figure 6: Demo screenshot before generation: the dialogue input area is visible together with enough surrounding UI context to show how a user starts the workflow.

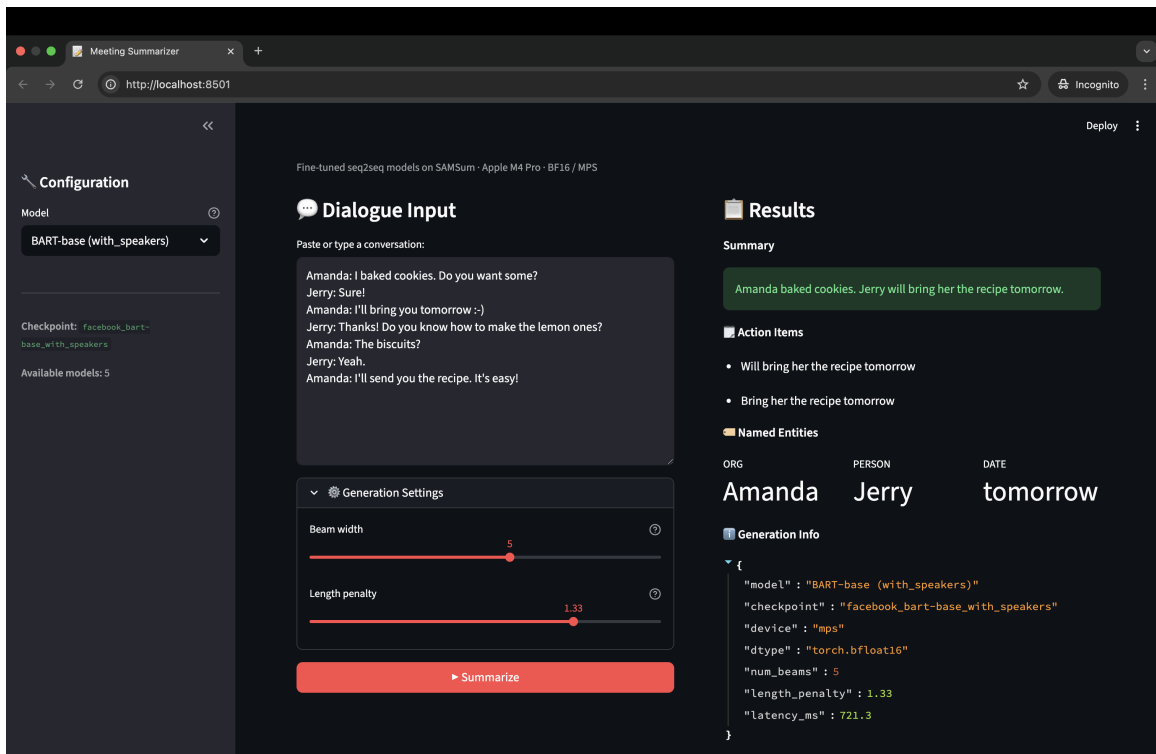


Figure 7: Demo screenshot after generation: the summary output is visible together with at least one structured output such as action items, named entities, or the latency/model information block.

10.3 Qualitative Demo Examples

Example	Input type	Expected visible output	Why it helps scoring
A	Short coordination dialogue	Faithful summary + correct speaker action	Shows baseline usability
B	Longer planning dialogue	Summary + action items + entities	Shows richer structured output

11 Reproducibility

11.1 Minimal Reproduction Path

```
# 1) Environment
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2) Data and preprocessing
python3 scripts/predownload_assets.py
python3 scripts/data_audit.py
python3 scripts/preprocess.py --variants with_speakers no_speakers split_speakers

# 3) Training and evaluation
python3 scripts/train.py
python3 scripts/decoding_ablation.py
python3 scripts/evaluate_faithfulness.py
```

```
python3 scripts/bootstrap_ci.py

# 4) Demo
bash scripts/run_app.sh
```

11.2 Critical Files

Path	Purpose
config.yaml	primary training and decoding configuration
config_extended.yaml	extended-schedule experiment (E8)
scripts/train.py	main fine-tuning pipeline
scripts/decoding_ablation.py	decoding sweep used for D27 selection
results/metrics/	all reported quantitative outputs
models/best/	best saved checkpoints
scripts/app.py	Streamlit demo application

All experiments use `seed=42`, and the report only cites metrics that are already committed in repository outputs.

12 Conclusion

- The project **met its success criterion**, achieving ROUGE-L 40.12 on the full SAMSum test split.
- The best system is **BART-base with speaker tags and tuned beam-search decoding**.
- The strongest empirical insight is that **speaker attribution is essential** for dialogue summarization quality.
- The main remaining weakness is **faithfulness**, especially hallucination and action-direction inversion.
- The project is **reproducible and demo-ready** on consumer Apple Silicon hardware.